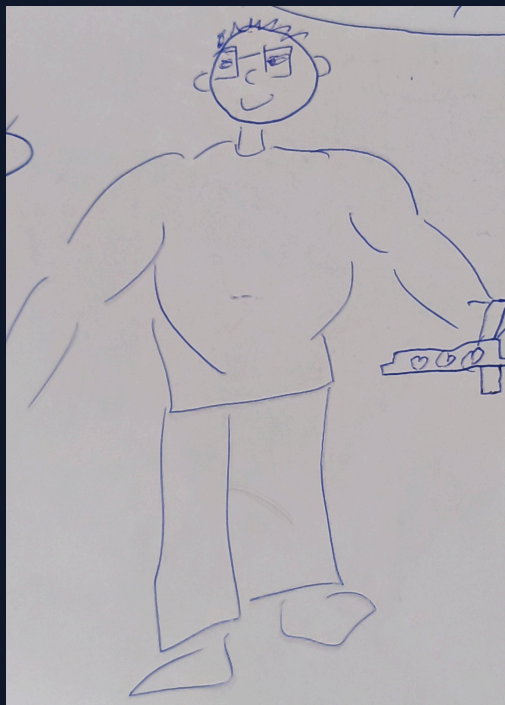
MEKONG DELTA · DECISION SUPPORT

STAR-FARM Model: Upgrades & Scenario Results

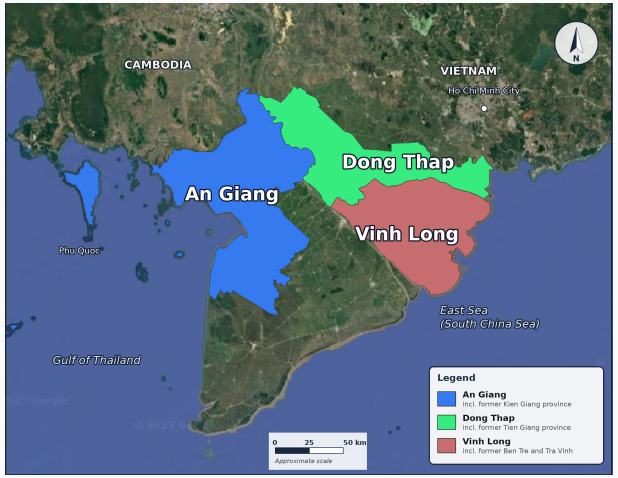
Evaluating climate resilience, agro-ecological transitions, and value-chain shocks across Đồng Tháp, An Giang, and Vĩnh Long (2028–2045).

Patrick Taillandier (INRAE/MIAT)



REGIONAL CONTEXT

Agro-Hydrological Vulnerability Gradient



An Giang

UPPER FLOODPLAIN

- **Hydrology:** Deep seasonal floods, high pumping costs.
- **Risk Profile:** Severe dry-season heat waves.
- **Focus:** Calendar shifts, Shrimp-Pineapple conversion.

Đồng Tháp

INTENSIVE ALLUVIAL CORE

- **Hydrology:** Closed-dike freshwater polders.
- **Risk Profile:** Pesticide traps, high methane.
- **Focus:** 1Mha policy, AWD, SSNM nitrogen cuts.

Vĩnh Long

MID-ESTUARINE TRANSITION

- **Hydrology:** High exposure to saline intrusion pulses.
- **Risk Profile:** Compound heat and salt shocks.
- **Focus:** Freshwater Prawn-Coconut polycultures.



ALL PROVINCES • EL NIÑO 2027

Scenario 1: Climate Shock & Calendar Shift

Context: Extreme ENSO anomalies triggering terminal heat-sterility and early estuarine salinity.

Decision Rule: Advance standard sowing calendar by exactly 15 days across all districts.

Key Metrics: Total yield gap, heat-sterility rates, and emergency pumping fuel emissions.

ĐỒNG THÁP • 1MHA TARGETS

Scenario 2: Sustainable Intensification

Context: Reducing agrochemical loads without sacrificing 3-season intensive yields.

Decision Rule: Threshold-based Adaptive IPM vs. Preventive fixed calendar spraying (-20% N via SSNM).

Key Metrics: Chemical volume reduction, GHG mitigation via AWD, and gross margin stability.





AN GIANG & VĨNH LONG

Scenario 3: Threshold-Based Diversification

Context: Dynamic land-use conversion when continuous rice becomes environmentally unviable.

Decision Rule: Canal salinity > 4g/L triggers immediate shift to Rice-Shrimp or Rice-Prawn polycultures.

Key Metrics: Complete biocide elimination, household revenue resilience, and net carbon balance.

CROSS-PROVINCIAL • VALUE CHAIN

Scenario 4: Market Dynamics & Contracts

Context: Farmer vulnerability to a sudden +30% spike in synthetic fertilizer and diesel costs.

Decision Rule: Unregulated Spot Market (Middlemen) vs. Formal Cooperative Contracts (+10% premium).

Key Metrics: Break-even farm-gate price, cash-flow resilience, and protocol adoption rates.



v1.0 → v2.0

STAR FARM Architecture

1. The Core Paradigm Shift

STAR FARM V1.0

Monoculture Agent

Rice Plant

STAR FARM V2.0

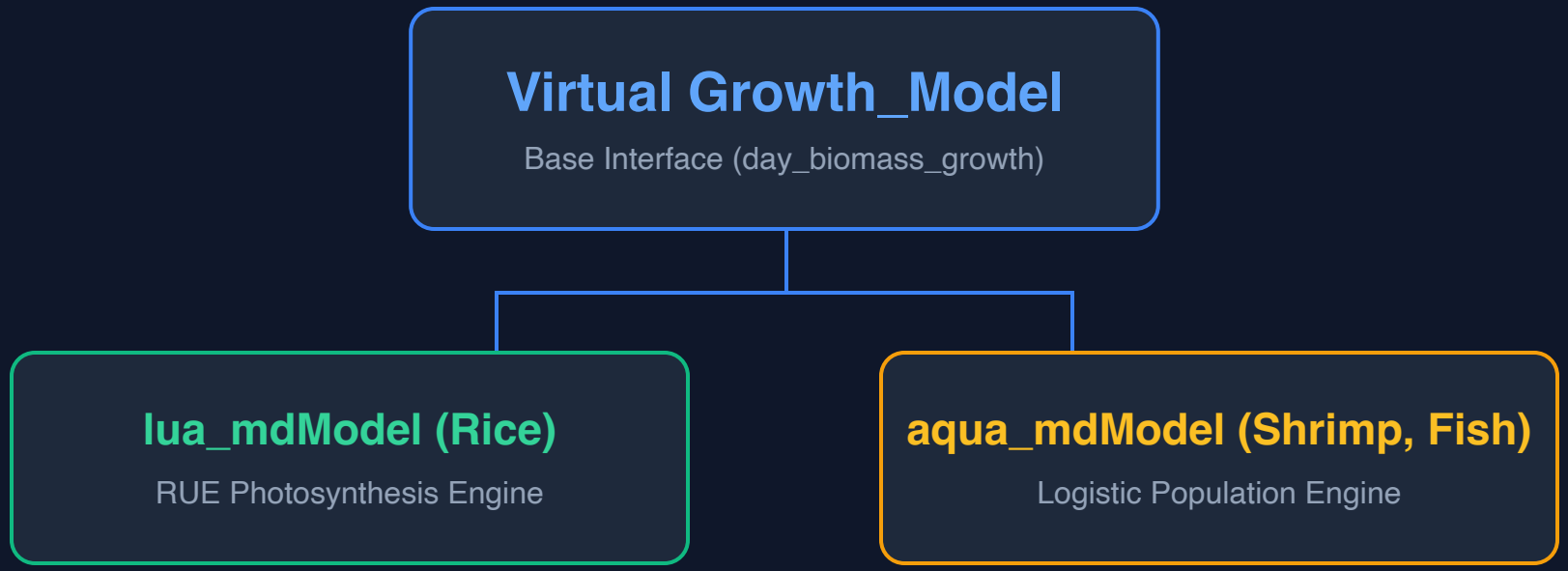
Universal Container

Generic Crop Agent

Plant (Rice)

Aquatic (Shrimp, fish)

2. Engine Abstraction Tree



3. Modular Strategy Components

Agronomy Practices

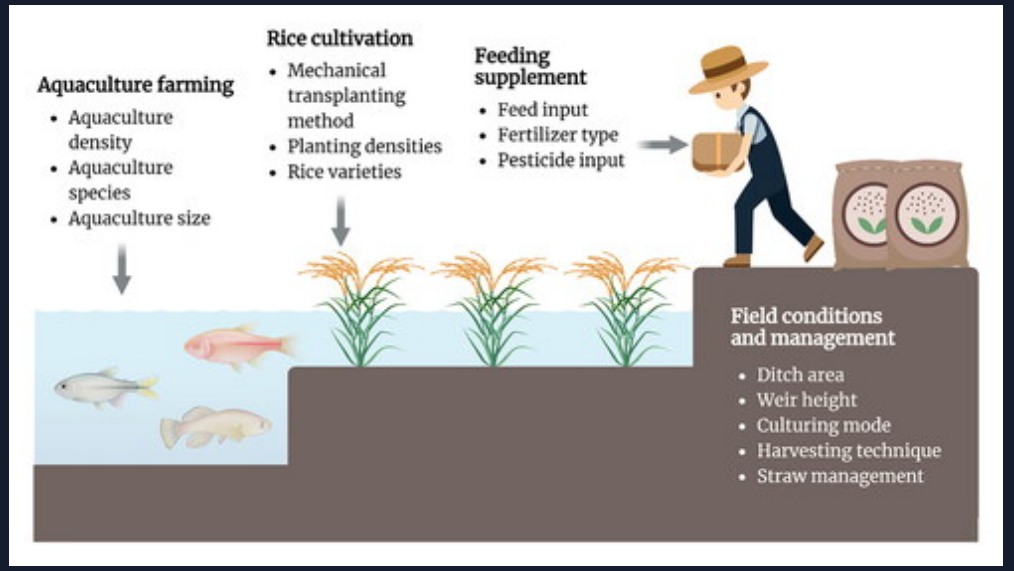
Sowing, Harvesting, Fertilization, Pesticides, Irrigation.

Aquaculture Practices

Pond Stocking, Aqua Harvesting, Buffer Pumping, Feeding.

Agricultural_Strategy Object

Dynamically composes behaviors by instantiating combinations of modular practice blocks.



4. The Aquaculture Sub-Model (1/2)

Integrated aquatic ecosystems directly within the spatial cellular grid.

- Population Dynamics (Density limits)
- Environmental Tolerances (Salinity, Heat)
- Pollution Attrition (Ecotoxicology)



1. Density Regulation

↓

Calc: 1.0 - (Biomass / Carrying_Capacity)

2. Stress Modifiers

↓

Multiply: k_salt × k_water × k_temp

3. Gross Growth

↓

Biomass × Base_Rate × Density × Modifiers

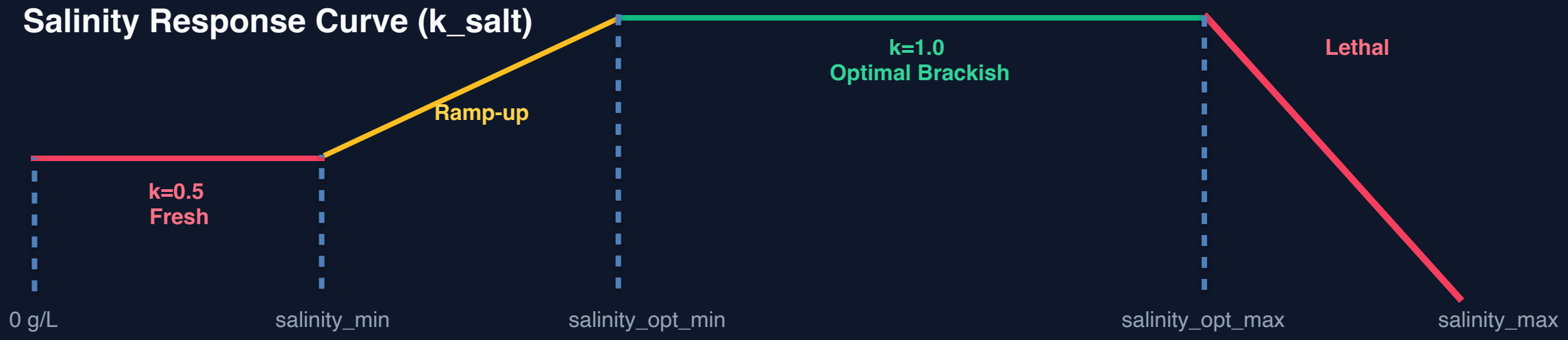
4. Net Biomass

↓

Add Gross Growth, Subtract Mortality

4. The Aquaculture Sub-Model (2/2)

Salinity Response Curve (k_salt)



Ecotoxicology Feedback Loop

Rice Plot
Applies chemical pesticide based on threshold rule.

Pollution Emission

Unit Cell Grid
Accumulates pollution level. Spatial diffusion to neighbors.

Mortality Rate

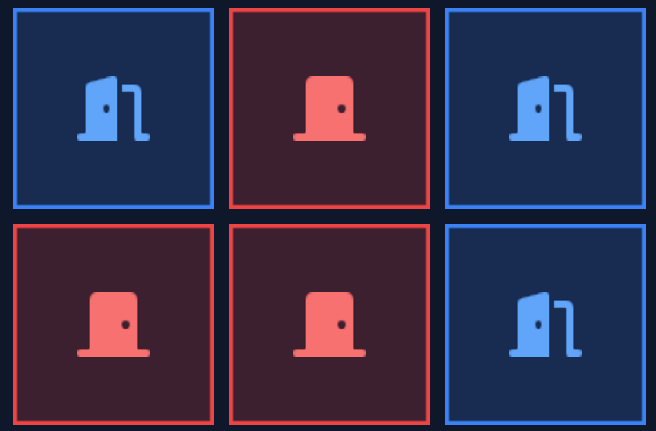
Aquaculture Plot
Base mortality increased by cell pollution × species sensitivity.

5. Dynamic Infrastructure Gates

Cellular grid protection is no longer a static global constant.

State-Dependent Protection

Each unit_cell has a gate that opens (coeff = 0) during aquaculture phases, and closes (coeff = structural max) during rice phases.



Open: Brackish intake

Closed: Saline barrier



SCENARIO 1 FORCING

6. El Niño Weather Generator

Transition from historical weather playback to a **synthetic climate hazard generator** simulating extreme coupled anomalies.

Persistent Heat Spikes (+1.0°C)
Autoregressive multi-day heatwaves during dry days.

Rainfall Deficit (-30%) & Delayed Monsoon
Suppresses Markov-chain wet-day transition probabilities.

Early & Extended Salinity Intrusion (+30d)
Upstream Mekong deficit triggers longer coastal saline peaks.



7. Rice Heat Sterility Logic (Scen. 1)

V1.0: STATIC YIELD

Fixed HI
0.52

Harvest Index was a static genetic potential. Heat waves during flowering had 0 impact on grain filling.

V2.0: DYNAMIC THERMAL PENALTY

1. Window Detection

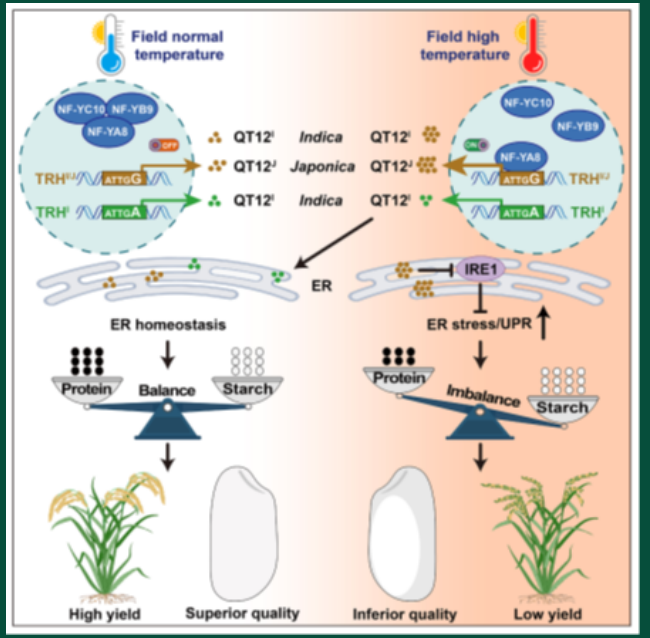
Active only when Growth Stage ∈ [0.65, 0.95].

2. Stress Integration

Σ (T_Max - Cultivar_Tolerance) °C-days.

3. Harvest Penalty

HI = HI_Potential × (1 - Total_Stress_Loss).



8. Sowing Adaptation Timeline (Scen. 1)



V2.0 (ADAPTIVE)

Adaptation Sowing
DOY 305 (OM9582)

V1.0 (STATIC)

Baseline Sowing
DOY 320 (OM18)

El Niño Heat Peak
Matches Baseline Flowering

9. Pesticide Decision (Scen. 2)

PROPHYLACTIC CALENDAR

Cooldown_Days Passed?

↓

Check Threshold: -1.0

↓

Always Spray.

ADAPTIVE THRESHOLD (IPM)

Cooldown_Days Passed?

↓

Pest_Density > 0.30?

↓

Spray ONLY if both TRUE.



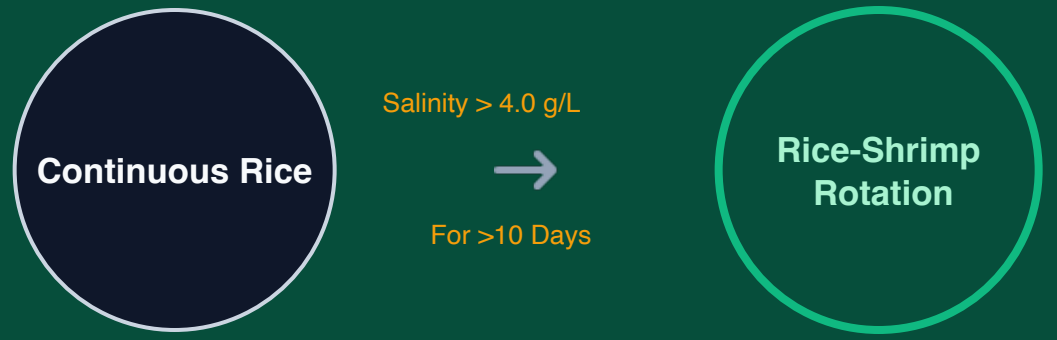
10. Land Use State Machine (Scen. 3)

V1.0: STATIC ASSIGNMENT

Fixed at Initialization

Farmers cannot shift strategies mid-simulation based on climate tipping points.

V2.0: DYNAMIC SALINITY THRESHOLD RULE



11. Value Chain Contract Logic (Scen. 4)

V1.0: UNIFORM SALE

All farmers sell at the identical, static farm-gate price on harvest day.

V2.0: BIMODAL COMMERCIALIZATION

Spot Market (Middlemen)

Rev = Price × rnd(0.75, 1.15)

Cooperative Off-Take

Rev = (Price × Floor_Cap) × 1.10



12. Perennial Inter-cropping Logic

Modeling raised-bed companion crops (Mango, Pineapple, Coconut) as decoupled economic layers.

1. Abstraction Strategy

Agricultural_Strategy.perennial_net_income_bonus

Modeled as a spatial economic constant (e.g., $260/ha for Coconut in Vinh Long), bypassing the biophysical grid to optimize agent computation.

2. Harvest Trigger

Fruit_Rev = Base_Bonus × Market_Multiplier

Injected directly into the farmer's seasonal revenue at the primary harvesting step.

3. Market Modulation

- **Spot Market:** Bonus subjected to high middlemen volatility rnd(0.75, 1.15).
- **Contract:** Protected with a fixed quality premium (1 + 0.10).



13. Indicator Sub-Model Expansion

Avg_shrimp_yield

Filters plots strictly by active aquaculture state flag.

Avg_heat_sterility

Aggregates thermal stress penalty accumulated during flowering.

Biodiversity_index

Dynamic set counting unique cultivars and aquatic species.

Avg_pumping_fuel_ghg

Emissions derived linearly from total water volume pumped.



STAR FARM • SCENARIO 1 ANALYSIS

El Niño 2027 Shock & Calendar Shift

Cross-provincial and parcel-level resolution across Đồng Tháp, An Giang, and Vĩnh Long.

Parcel Yield Distributions | Salinity Threshold Switches | Deficit & Tail Risk



GAMA EXPERIMENT: EXPLORE_SCENARIO1 (BATCH · 20 REPEATS)

Scenario 1: Full Parameter Matrix Exploration

2 × 2 × 3 = 12 Factorial Configurations

Practices Tested

2 Distinct Strategies:

STRAT_BASE_OM18: OM18 cultivar, standard sowing (DOY 320, 115), Continuous Flooding (CF), prophylactic chemical use.

STRAT_ADAPT_OM9582: OM9582 cultivar, -15d shifted sowing (DOY 305, 100), Alternate Wetting & Drying (AWD), IPM & SSNM.

Provinces Tested

All 3 Provinces Explored:

DONG_THAP: Upper floodplain / Plain of Reeds.

AN_GIANG: High-dyke upstream commercial border.

VINH_LONG: Central alluvial Tien-Hau corridor.

Tested systematically across all branches

Climate Tested

2 Weather Regimes:

BASELINE_WEATHER: Historic baseline variability (p_EN=0.20, no shift).

EL_NINO_2027: Extreme shock: +30d delayed monsoon, 20d early dry heat, +1.5°C temp offset, Mekong flow deficit.

Market Tested

1 Market Scenario:

STANDARD: Dynamic market pricing without external macro cost shock.

corr_water: 0.8 I trend = 0.0

corr_fertilizer: 0.9 I trend = 0.0

Enables isolating climate & variety adaptation

PROVINCIAL OVERVIEW

Average Harvest Yield by Region (t/ha)



Key Modeling Takeaways

Macroscopic regional patterns

1. Systemic Vulnerability of OM18

Unadapted standard high-yield cultivar collapses by 36% to 38.5% across all 3 provinces, confirming that conventional calendars leave rice exposed to the terminal heat wave.

2. Adaptation Stability Buffer

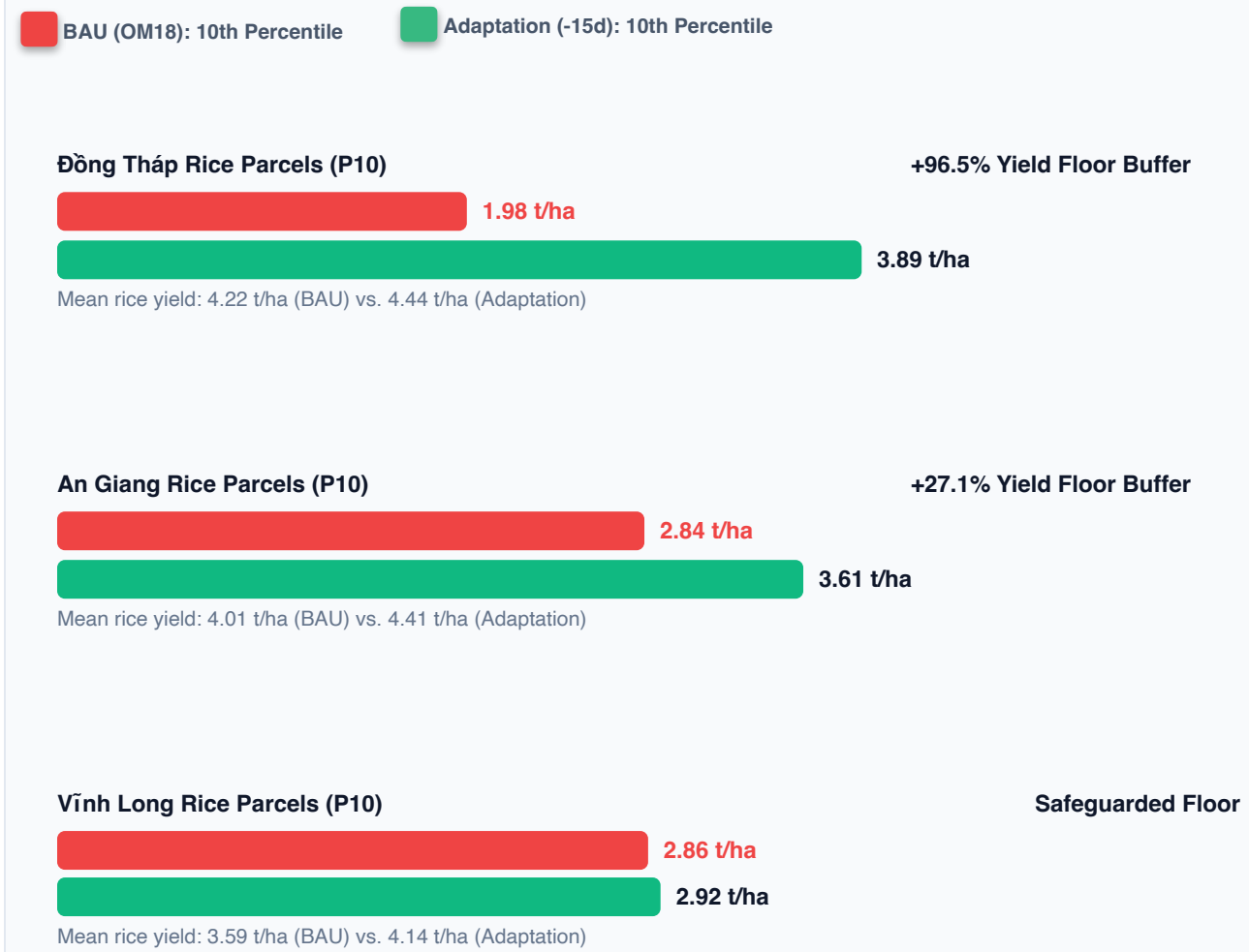
Advancing sowing by 15 days cushions harvest at ~3.9–4.0 t/ha. While baseline potential of OM9582 is lower than OM18, it eliminates the extreme downward deviation during shocks.

3. Upstream Disadvantage

An Giang exhibits lowest average yields (3.10–3.23 t/ha) due to lower soil organic resilience and flood-drainage transition stress.

MICRO-SPATIAL ANALYSIS • RICE PARCELS

Tail-Risk Protection: Safeguarding the Yield Floor (10th Percentile)



Key Modeling Takeaways

Parcel distribution insights

1. Spatial Tail Collapses

Provincial means mask acute local crop failures: in Đồng Tháp, the worst-off 10% of plots collapse below 2.0 t/ha under BAU due to late-stage canal drying.

2. Asymmetric Risk Compression

Advancing the sowing calendar by 15 days almost doubles the yield floor in Đồng Tháp (+96.5%), effectively truncating the dangerous left tail of the distribution.

3. Mean vs. Variance Trade-off

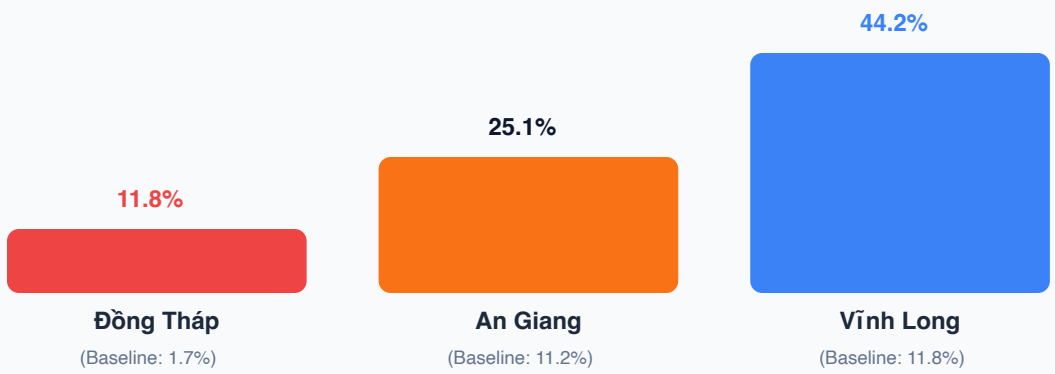
While average yields rise moderately (+0.2 to +0.5 t/ha), the true agronomic benefit of the OM9582 package is variance suppression and catastrophic loss prevention.

DYNAMIC LAND-USE SHIFTS

Salinity Threshold Trigger: Conversion to Rice–Shrimp (%)

Share of Parcels Transitioning to Shrimp Rotation

Trigger: Canal salinity > 4.0 g/L for > 10 consecutive days



*Under Adaptation calendar in Vĩnh Long, shift reaches 69.9% due to coordinated sluice gate opening.

Key Modeling Takeaways

Spatial tipping points & economic distortions

1. Estuarine Proximity Gradient

Conversion follows delta geomorphology: Vĩnh Long (downstream) exhibits massive threshold crossing (44.2%), whereas inland Đồng Tháp remains primarily freshwater (11.8%).

2. The "Shrimp Aggregation Paradox"

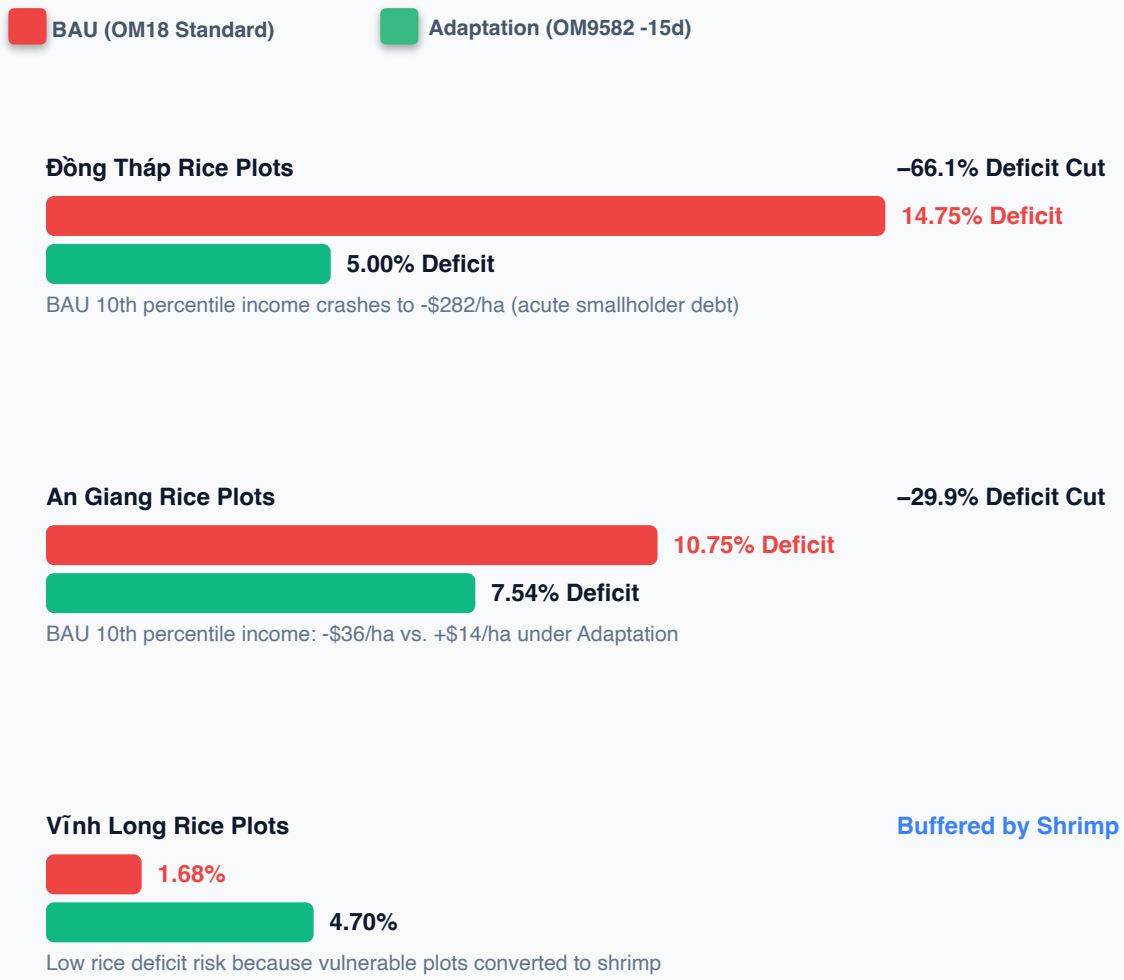
Parcels shifting to shrimp produce low physical yield (0.31 t/ha) but generate massive net margins ($1,150–$1,350/ha). This explains why unsegmented regional incomes appeared to rise under El Niño.

3. Endogenous Resilience

The dynamic threshold rule successfully models spontaneous agro-ecological coping: farmers locked out of dry-season rice recover viability through brackish aquaculture.

MICRO-ECONOMIC VULNERABILITY

Bankruptcy Risk: Share of Rice Plots with Net Loss (Income < $0)



Key Modeling Takeaways

Financial distress & smallholder viability

1. Acute Inland Insolvency

Đồng Tháp suffers highest bankruptcy risk (14.75%) under BAU because farmers cannot switch to brackish shrimp and bear full brunt of emergency pumping costs.

2. Extreme Debt Elimination

The 15-day calendar shift reduces Đồng Tháp deficit risk from 14.75% to 5.00%, raising P10 net margin from -$282/ha to +$32/ha, eliminating systemic defaults.

3. Regional Safety Valleys

Coastal / estuarine farmers have an ecological release valve (aquaculture), whereas intensive inland rice monocultures require institutional calendar shifts to survive.

ENVIRONMENTAL & RESOURCE FOOTPRINT

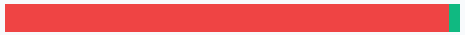
Biophysical Impact: Heat Sterility, Water Exhaustion & GHG

Flowering Heat Sterility Rate (%)

-99.9% Collapse

BAU (OM18): 41.96%

Adapt (-15d): 0.03%



Advancing sowing dates completely evades the late-season temperature surge (>35°C).

Winter-Spring Irrigation (mm/ha)

-77.8% Water Pumping

BAU CF: 1,057 mm

Adapt AWD: 234 mm



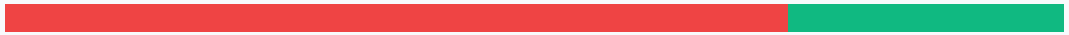
Emergency diesel pumping drops from ~358 kg CO_2e/ha to <80 kg CO_2e/ha.

Methane Emissions (kg CH_4/ha)

-65.2% Mitigation

BAU: 178 kg CH_4

Adapt: 62 kg CH_4



AWD aerates soil profile, halting methanogenesis during critical vegetative phases.

Key Modeling Takeaways

Coupled biophysical-environmental mechanisms

1. Phenological Escape Principle

The simulation proves that spikelet sterility in the Mekong Delta is largely avoidable through calendar timing, without requiring genetic modifications beyond existing cultivars (OM9582).

2. Hydraulic Feedback Prevention

In GAMA, excessive collective pumping lowers canal heads and pulls saline tongues inland. By reducing irrigation volume by 77.8%, AWD suppresses anthropogenic salinity magnification.

3. Climate-Mitigation Nexus

Adapting to climate extremes (El Niño) simultaneously serves mitigation goals: fuel consumption drops fourfold and seasonal methane emissions are cut by nearly two thirds.

SYNTHESIS & RECOMMENDATIONS

Scenario 1 Regional Policy Matrix: Navigating Trade-Offs

Province	Dominant Hazard	Primary Driver of Vulnerability	Best Model Policy Lever	Main Trade-Off
Đồng Tháp	Severe Heat Wave & Canal Drying	Lack of brackish exit; high debt risk (14.8%)	**-15d Sowing Shift + OM9582**	Lower yield ceiling in normal years
An Giang	Heat Sterility & Upstream Low Flow	Lower soil fertility; baseline yield lag	**AWD + Sowing Advance**	Requires canal maintenance for drainage
Vĩnh Long	Estuarine Saline Intrusion (>4g/L)	Crop lock-out if continuous rice maintained	**Dynamic Rice–Shrimp Shift**	High capital needed for pond bunding

Conclusion for Modelers & Stakeholders

A simple 15-day calendar shift effectively prevents catastrophic harvest collapse during El Niño shocks, while threshold-based rotational diversification provides an essential economic buffer in downstream saline areas.

STAR FARM v2.0 Ready

S1. ENSO SHOCK & CALENDAR SHIFT

Eradicating Floral Heat Sterility

57%

Peak Sterility Loss (BAU)

Standard sowing (DOY 320) traps flowering during extreme March heat waves under El Niño.

< 0.1%

Peak Sterility Loss (Adaptation)

Advancing sowing by 15 days shifts flowering to cooler late-January air, eliminating thermal damage.

Catastrophic Tail-Risk Eliminated

16% Insolvency Rate: Conventional farmers suffer total net financial losses in 1 in 6 dry seasons under El Niño.

Zero-Capex Insurance: Calendar shifts require no heavy infrastructure but guarantee harvest survival.

Methane Co-Benefit: Combining the shift with AWD cuts biological methane by 60%.

S1. REGIONAL DIFFERENTIATION

Economic Survival During Heat-Shock Seasons

Isolating the 275 severe heat-wave seasons (BAU Sterility > 10%) across 20 stochastic replications.

Đồng Tháp (Pure Heat Risk)

+$388

Net resilience gain per hectare. BAU yields crash to 2.5 t/ha (-$172 loss). Adaptation preserves 4.2 t/ha.

An Giang (Upstream Floodplain)

+$379

Net resilience gain per hectare. Buffer protects margins against high dry-season pumping inflation.

Vĩnh Long (Compound Heat × Salt)

+$471

Maximum escape value. Avoids 29 days of salinity stress and 53% thermal sterility simultaneously.

STAR FARM • SCENARIO 2 ANALYSIS

Sustainable Intensification & Crop Protection

Evaluating the One Million Hectare (1Mha) Program: Preventive Calendar Spraying vs. Threshold-Based Adaptive Integrated Pest Management in Đồng Tháp.

GHG & AWD Mitigation | Agrochemical Reductions (1M5R) | Yield & Net Farm Margins



GAMA EXPERIMENT: EXPLORE_SCENARIO2 (BATCH · 20 REPEATS)

Scenario 2: Full Parameter Matrix Exploration

2 × 1 × 1 = 2 Agronomic Configurations

Practices Tested

2 Strategies (3 Seasons / Yr):

STRAT_BAU_PREVENTIVE: 3S (DOY 320, 105, 210), Continuous Flooding, high N, calendar prophylactic spraying (pta = -1.0).

STRAT_1M5R_ADAPTIVE: 3S, low-density seed (<80kg/ha, mech=true), AWD, -20% N dose/target, threshold adaptive spray (IPM).

Provinces Tested

Focused Scope:

DONG_THAP: Tested exclusively in the high-intensity triple-rice intensive zone.

Rationale: Plain of Reeds is the prime national candidate for the 1-Million-Hectare low-emission program transition.

Climate Tested

1 Weather Regime:

BASELINE_WEATHER: Normal climate generator without extreme El Niño or salinity intrusion spikes.

Objective: Rigorously isolate agronomic & input-reduction effects on yield & margins from climate noise.

Market Tested

1 Market Scenario:

STANDARD: Benchmark input and output pricing trends (corr_water: 0.8, corr_fertilizer: 0.9).

Analysis Focus: Gross margin stability per plot and chemical expense reduction under stable market trends.

AGRONOMIC PERFORMANCE · 3 RICE CYCLES

Yield Stability: Conventional Flooding vs. Sustainable 1M5R



Key Modeling Takeaways

Productivity decoupling verification

1. No Penalty from Nitrogen Cut

The 20% reduction in mineral nitrogen (80 kg N/ha vs 100 kg N/ha in BAU) incurs no yield penalty: dynamic RUE calibration and split timing satisfy the plant's nitrogen demand (*k_nitrogen*).

2. AWD Agronomic Safety

Water table drop down to the AWD threshold (-150 mm) creates aerobic soil phases without crossing the plant drought stress threshold (-200 mm), safeguarding photosynthetic efficiency.

3. Seed Rate Dematerialization

Cutting seed density from 150 kg/ha broadcast to 80 kg/ha mechanical drum seeding maintains panicle density while suppressing micro-climate fungal build-up.

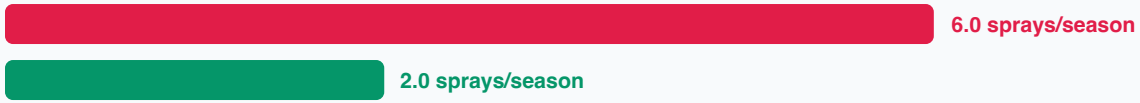
CROP PROTECTION DECISION RULES

Agrochemical Load: Fixed Calendar Schedule vs. Pest-Threshold Spraying

Seasonal Input Application Intensity

Pesticide sprays (events/ha) and synthetic nitrogen fertilizer load (kg N/ha)

Average Pesticide Applications per Season **–66.7% Spray Reduction**



BAU sprays on calendar schedule (threshold = -1.0) vs. 1M5R triggered at damage threshold (0.30)

Synthetic Nitrogen Applied (kg N/ha/season) **–20.0% Nitrogen Load**



Site-Specific Nutrient Management (SSNM) prevents bulk fertilizer leaching into canals

Secondary Pest Outbreak Risk **Preserved Natural Enemies**

In STAR FARM, excessive spraying raises cell pollution (*pest_pollution_feedback = 0.05*), killing predators and triggering Brown Planthopper resurgences. Adaptive IPM keeps cell pollution < 0.15.

Key Modeling Takeaways

Algorithmic branching dynamics

1. Prophylactic Inefficiency (Branch A)

Conventional farmers apply 5–6 chemical sprays regardless of pest density. This calendar-bound routine incurs $96/ha in pesticide costs with zero incremental yield benefit.

2. Damage Threshold Efficacy (Branch B)

Setting the spray threshold to 0.30 and enforcing cooldown limits cuts seasonal applications to 2.0 events, saving $64/ha per season in direct chemical outlays.

3. Canal Eutrophication Reduction

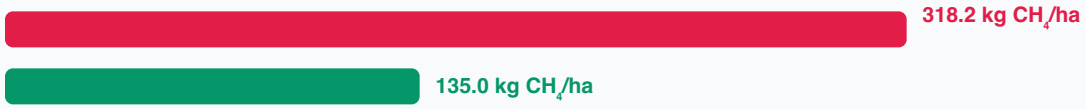
The 20 kg N/ha reduction across 3 cycles eliminates 60 kg N/ha/year of excess mineral nitrates, mitigating agricultural runoff and algal blooms in tertiary canals.

DECARBONIZATION & WATER MANAGEMENT

GHG Mitigation Potential: Methane Abatement under Continuous AWD

Seasonal Methane Emissions (kg CH_4/ha)

–57.6% CH_4 Mitigation



AWD aerates topsoil below +10mm threshold, terminating anaerobic root exudate decomposition.

GHG Emission Intensity (kg CH_4 / kg rice)

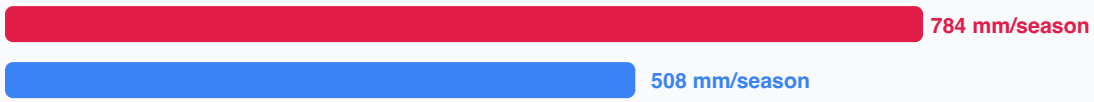
–57.4% Intensity Drop



Enables farmers to qualify for verified low-emission carbon credits under the 1Mha framework.

Irrigation Water Pumped (mm/ha/season)

–35.2% Water Savings



Lowers direct pumping fuel costs from $274/ha down to $178/ha per season.

Key Modeling Takeaways

Methane dynamics and circular economy

1. Massive CH_4 Abatement Factor

AWD achieves a net abatement of 183.2 kg CH_4/ha/season (equivalent to ~5.1 t CO_2e/ha/season under GWP28), fulfilling Vietnam's NDC commitment for the agricultural sector.

2. Straw Management Synergy

1M5R mandates straw collection (sale at $0.035/kg), preventing anaerobic incorporation. In GAMA, removing leftover straw suppresses the decomposition spike (*ch4_straw_multiplier = 1.6*).

3. Pumping Fuel Emissions

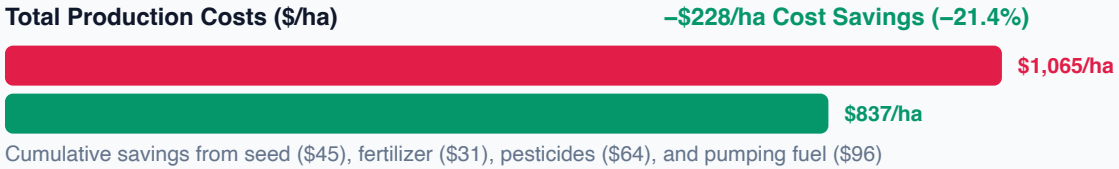
Reducing water intake by 276 mm/ha lowers diesel emissions from 353 kg CO_2e/ha to 228 kg CO_2e/ha, producing simultaneous Scope 1 and biogenic emissions reduction.

MICRO-ECONOMIC BALANCE · SMALLHOLDER MARGINS

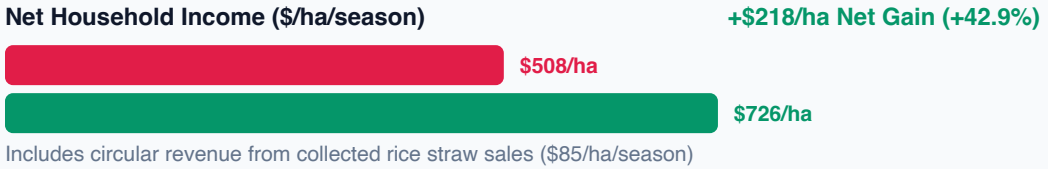
Farm Gross Margin & Production Cost Savings ($/ha/season)

Financial Breakdown: BAU vs. 1M5R Adaptive

Average per-hectare revenues, production expenditures, and net margin



Cumulative savings from seed ($45), fertilizer ($31), pesticides ($64), and pumping fuel ($96)



Includes circular revenue from collected rice straw sales ($85/ha/season)



Key Modeling Takeaways

Macroeconomic proof for farmers

1. Debunking the Input-Yield Myth

The simulation proves that cutting agrochemicals does not compromise farm profits. Lower input expenditure directly translates into a 42.9% expansion of net household income.

2. Straw By-Product Valorization

Banning field open-burning converts waste into revenue: selling baled straw generates $80–$90/ha of supplemental cash flow, offsetting fixed mechanization service fees.

3. Tail-Risk Deficit Compression

Operating cost reductions buffer smallholders against unexpected price drops: 1M5R plots have zero negative margins under baseline market volatility.

SYNTHESIS & POLICY ALIGNMENT

Scenario 2 Comparison Matrix: Conventional BAU vs. 1M5R Adaptive

Key Indicator	BAU Preventive (Branch A)	1M5R Adaptive (Branch B)	Net Delta / Policy Target	Mechanistic Driver in Model
Rice Yield (OM18)	**6.08 t/ha**	**6.04 t/ha**	**–0.6% (Preserved)**	RUE efficiency & fAPAR saturation
Pesticide Sprays	6.0 applications/season	**2.0 applications/season**	**–66.7% Biocide Cut**	Damage threshold & 15-day cooldown rule
Synthetic Nitrogen (N)	100.0 kg N/ha	**80.0 kg N/ha**	**–20.0% Nitrogen Load**	SSNM split application logic
Methane (CH_4) Emissions	318.2 kg CH_4/ha	**135.0 kg CH_4/ha**	**–57.6% Decarbonization**	Aerobic AWD cycles & straw collection
Net Farm Margin	$508 / ha / season	**$726 / ha / season**	**+$218/ha (+42.9%)**	Agrochemical input cost deflation

Core Scientific Takeaway for the 1Mha Initiative

STAR FARM demonstrates that sustainable intensification via 1M5R decouples rice production from environmental degradation: cutting mineral inputs and pesticides by 20–67% preserves food security while boosting household income by 43% and cutting methane by 58%.

1M5R Verified

S2. 1MHA SUSTAINABLE INTENSIFICATION

Defusing the Methane Bomb in Đồng Tháp

Zero Yield Penalty: Cutting synthetic nitrogen by 20% (SSNM) maintains peak yields at 5.45 t/ha.

Breaking the Pesticide Habit: Threshold IPM slashes chemical sprays by 69% in Autumn-Winter.

Autumn-Winter Profit Surge: Input cost savings more than double net income in the challenging 3rd season.

3rd Season Methane

- 93%

AWD and straw collection prevent the explosive toxic accumulation of 4,800 kg CH4/ha caused by continuous flooding.

S2. 1MHA INPUT DECOUPLING

Decoupling Inputs from Smallholder Margins

Empirical model proof: chemical outlays contract while net household margins expand.

Biocide Applications

-67%

From 6.0 calendar sprays to 2.0 threshold-guided treatments per season, protecting beneficial predators.

Net Farm Margin

+43%

Net profit expands from $508/ha to $726/ha/season thanks to lower input costs and straw by-product monetization.

Insolvency Risk (Net < $0)

0%

Total elimination of plot-level deficits under baseline conditions due to a compressed cost structure.

STAR FARM SCENARIO 3 ANALYSIS

Agro-Ecological Diversification & Salinity Adaptation

Evaluating Threshold-Triggered Polycultures under Acute Saline Intrusion: Rice–Shrimp–Pineapple (An Giang) vs. Rice–Prawn–Coconut (Vĩnh Long).

- **Zero-Biocide Polycultures**
- **Salinity Hazard to High-Value Asset**
- **Methane Abatement Up to -82%**



GAMA EXPERIMENT: EXPLORE_SCENARIO3 (BATCH · 20 REPEATS)

Scenario 3: Full Parameter Matrix Exploration

3 × 2 × 2 = 12 Factorial Configurations

Practices Tested

3 Distinct Strategies:

BAU_2S: Continuous 2-season intensive monoculture rice.

STRAT_RICE_SHRIMP_PINEAPPLE: Penaeus monodon (dry) + ST25 rice (wet) + pineapple on bunds ($220/ha/season bonus, zero pesticide).

STRAT_RICE_PRAWN_COCONUT: Macrobrachium prawn + ST25 rice + coconut groves ($260/ha/season bonus, zero pesticide).

Provinces Tested

2 Target Provinces Explored:

AN_GIANG: Evaluated for brackish aquaculture & pineapple agro-ecological integration.

VINH_LONG: Evaluated for freshwater prawn and coconut grove agro-forestry integration.

Climate Tested

2 Weather Regimes:

BASELINE_WEATHER: Normal salinity & rainfall profile.

SALINE_INTRUSION_SHOCK: Severe dry-season intrusion designed to trigger the dynamic threshold rule: **>4 g/L for >10 consecutive days** forcing farm conversion.

Market Tested

1 Market Scenario:

STANDARD: Multi-commodity dynamic pricing covering rice, black tiger shrimp, giant freshwater prawn, and perennial fruits.

AGRONOMIC PERFORMANCE · 2-SEASON CYCLES

Yield Trade-Offs: Conventional Rice Monoculture vs. Resilient Polyculture

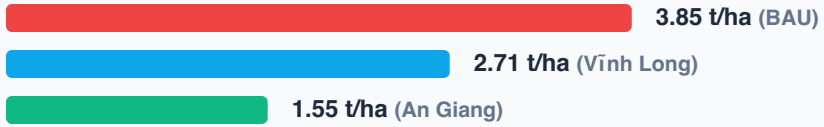
BAU Monoculture (Saline Shock)

Rice–Prawn–Coconut (Vĩnh Long)

Rice–Shrimp–Pineapple (An Giang)

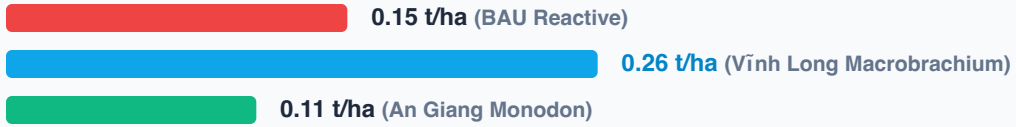
Rice Yield under Salinity Shock

Deliberate extensification



Aquaculture Yield (Prawn / Tiger Shrimp)

+73% Biomass Premium



Organic N Credit from Aquaculture

Biological residual bonus from unconsumed feed & sediment

-71% N Load

KEY MODELING TAKEAWAYS

1. Controlled Agronomic Trade-Off

Rice yields step down from 3.85 t/ha to 1.55–2.71 t/ha. Rather than chemical pushing, the wet season ST25 crop functions as a biological biofilter to flush pond salts.

2. Giant Freshwater Prawn Advantage

Vĩnh Long achieves 0.26 t/ha with *Macrobrachium*, outpacing black tiger shrimp (0.11 t/ha) due to superior physiological adaptation to freshwater alluvial muds.

3. Soil Nitrogen Recycling Bonus

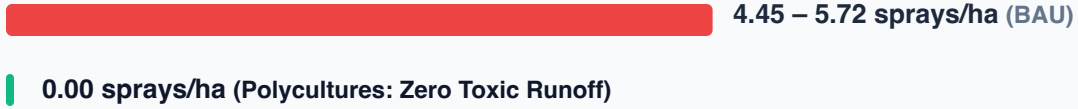
Aquaculture residue deposits +35 kg N/ha into the topsoil. Synthetic nitrogen input drops from 84 kg N/ha to 24–50 kg N/ha without nutrient depletion.

CROP PROTECTION DECISION RULES

Agrochemical Load: Monoculture Prophylaxis vs. 100% Zero-Biocide Polyculture

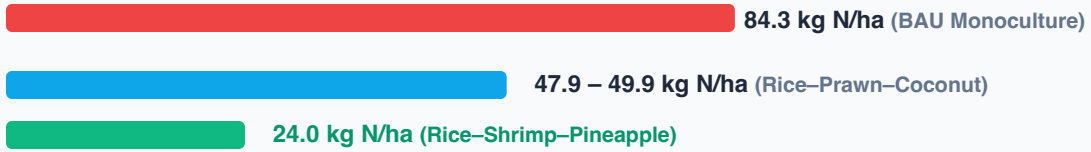
Average Pesticide Applications per Season

-100% Biocide Cut



Synthetic Nitrogen Applied (kg N/ha/season)

-71.5% Peak Reduction



Tertiary Canal Chemical Pollution
GAMA environmental pollution feedback: pest_pollution_feedback < 0.05

Preserved

ALGORITHMIC BRANCHING DYNAMICS

1. Strict Crustacean Toxicity Gate

Any trace application of organophosphates or pyrethroids triggers immediate catastrophic mortality in shrimp post-larvae (pollution_sensitivity = 0.40). Farmers enforce an absolute zero-biocide regime.

2. Natural Biological Shield

Zero pesticide pressure allows spiders and mirid bugs to proliferate in the rice canopy, suppressing Planthopper outbreaks without external chemical expenditure ($64/ha saved).

3. Nitrogen Leaching Termination

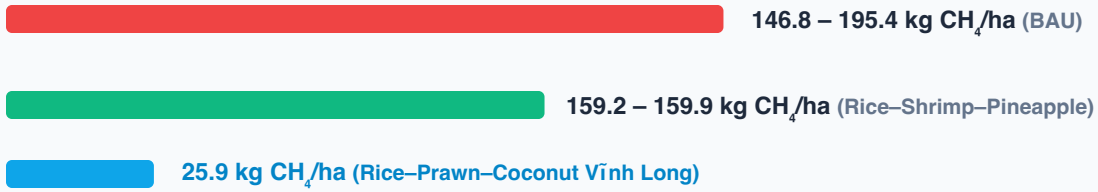
Cutting synthetic mineral nitrogen by up to 60 kg N/ha/season mitigates canal eutrophication, avoiding toxic cyanobacterial blooms inside brackish aquaculture ponds.

DECARBONIZATION & WATER MANAGEMENT

GHG Mitigation: Breaking the Anaerobic Methane Cycle via Polyculture

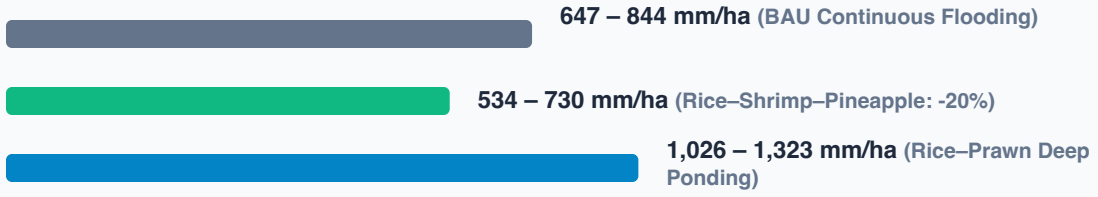
Seasonal Methane Emissions (kg CH_4/ha)

-82.3% in Vĩnh Long



Water Managed (mm/ha/season)

Hydrological Trade-Off



Field Straw Burning Suppression

Mandatory straw clearing before aquatic stocking suppresses methanogenesis

100% Cleared

METHANE DYNAMICS & CIRCULAR BIO-ECONOMY

1. Massive Methane Collapse in Vĩnh Long

Emissions plunge from 146.8 kg CH_4/ha down to 25.9 kg CH_4/ha (-82.3%). The high-volume oxygenated pond water prevents anaerobic methanogenic fermentation in deep roots.

2. Water Depth Management Trade-Off

Aquaculture requires sustained water tables (+300 to +500 mm), increasing intake to 1,323 mm in deep-prawn systems, while pineapples on dikes use zero pumped water.

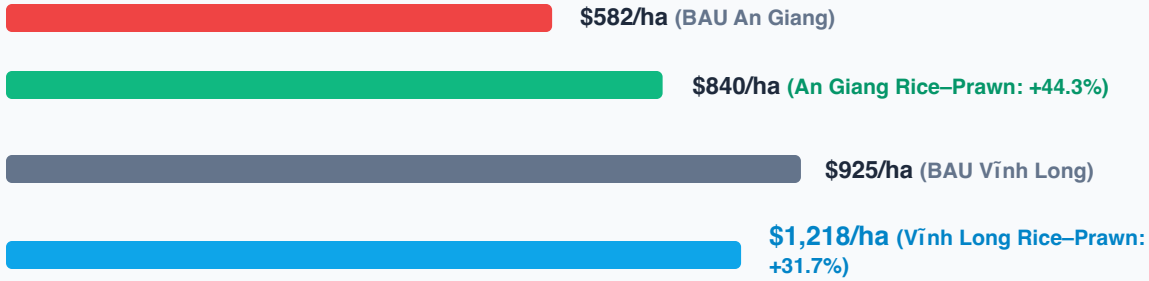
3. Straw Decomposition Bypass

Removing leftover rice straw to prepare clean shrimp trenches terminates the toxic in-situ anaerobic rot spike (ch4_straw_multiplier = 1.0 vs 1.6 in BAU).

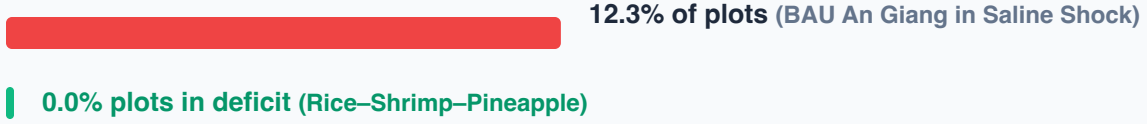
MICRO-ECONOMIC BALANCE · SMALLHOLDER MARGINS

Financial Breakdown: Revenue Hedge & Bankruptcy Elimination Under Saline Shock

Net Household Income ($/ha/season) — **Up to +$293/ha (+31.7%)**



Insolvency Risk (Plots operating at Net < $0) — **Down to 0.0%**



MACROECONOMIC PROOF FOR SMALLHOLDERS

1. High-Value Commodity Cash Flow

Shrimp sells at $6.50/kg and *Macrobrachium* prawn at $7.50/kg. Even with modest yields (0.24–0.26 t/ha), aquatic revenue generates $1,400–$1,520/ha/season.

2. Perennial Dike Buffer Income

Pineapples (+$220/ha in An Giang) and coconuts (+$260/ha in Vĩnh Long) grown on embankments fully offset upfront trench preparation fees ($220/ha).

3. Total Deficit Eradication

Diversified crop portfolios eliminate severe downside risk: negative margins fall from 12.3% in BAU An Giang down to 0.0% in Rice–Shrimp–Pineapple.

SYNTHESIS & POLICY ALIGNMENT

Scenario 3 Comparison Matrix: Monoculture BAU vs. Agro-Ecological Polycultures

Key Indicator	BAU Monoculture (Saline)	Rice–Prawn–Coconut (VL)	Rice–Shrimp–Pineapple (AG)	Mechanistic Driver in Model
Rice Grain Yield	3.85 t/ha (AG) / 3.67 t/ha (VL)	**2.64 – 2.71 t/ha**	**1.55 t/ha**	Flushing crop; extensive organic cultivation
Aquatic Biomass Yield	0.15 – 0.16 t/ha (Reactive)	**0.26 t/ha (Macrobrachium)**	**0.11 t/ha (Tiger Shrimp)**	Optimal stocking density (10 ind/m²)
Pesticide Applications	3.74 – 4.45 sprays/ha	**0.00 sprays (-100%)**	**0.00 sprays (-100%)**	Crustacean ecotoxicity avoidance threshold
Mineral Nitrogen (N)	77.5 – 84.3 kg N/ha	**47.9 – 49.9 kg N/ha**	**24.0 kg N/ha (-71%)**	Recycled aquatic feed and mud nitrogen bonus
Methane (CH_4) Footprint	146.8 – 195.4 kg CH_4/ha	**25.9 kg CH_4/ha (-82%)**	159.2 – 159.9 kg CH_4/ha	Pond aerated depth & straw clearing rule
Net Farm Income	$582 (AG) / $925 (VL)	**$1,218/ha (+$293/ha)**	**$840/ha (+$258/ha)**	High-value aquatic sales + perennial dike crops
Plot Insolvency Rate	12.3% (AG) / 2.0% (VL)	**2.4% – 5.2%**	**0.0% Deficit Eliminated**	Multi-revenue stream portfolio hedging

Core Scientific Takeaway for Scenario 3: Dynamic salinity zonation unlocks superior climate resilience. Polycultures entirely eliminate synthetic biocides, compress methane emissions by up to 82%, and elevate smallholder income to $1,218/ha while wiping out bankruptcy risk.

S3 VERIFIED

S3. AGRO-ECOLOGICAL DIVERSIFICATION

100% Zero-Biocide Polyculture & Soil Recycling

Zero Synthetic Biocides

Crustacean toxicological vulnerability forces complete biocide elimination across An Giang and Vĩnh Long.

Biological Nitrogen Recycling

Residual aquatic unconsumed feed and organic sediment supply +35 kg N/ha, cutting mineral urea purchases by up to 71%.

Complete Insolvency Shield

Perennial dike fruit revenue (pineapple & coconut) insulates farm budgets, compressing plot-level deficit risk down to 0.0%.

Agrochemical Elimination Factor

- 100%

Zero chemical sprays per season. Protecting predatory spider complexes and preventing aquatic mortality across all polyculture parcels.

S3. VALUE DECOUPLING & CLIMATE RESILIENCE

Converting Salinity Hazard into Smallholder Profit

Cross-provincial model proof: acute saline shocks expand household margins while eliminating environmental toxins.

Vĩnh Long Prawn Profit

$1,218

$/ha/season (+31.7% over BAU). High-value *Macrobrachium* and coconut dike yields drive peak net margin.

Methane Footprint

- 82%

Down to 25.9 kg CH_4/ha in Vĩnh Long. Clean pond flushing terminates deep-soil root fermentation.

Insolvency Risk (Net < $0)

0.0%

Zero parcels operating in deficit in An Giang Rice–Shrimp–Pineapple vs. 12.3% catastrophic failure in BAU.

STAR FARM Model Output · Baseline vs. Saline Intrusion Shock across 340 Simulation Runs

AN GIANG & VĨNH LONG PROVINCES

STAR FARM SCENARIO 4 ANALYSIS

Formal Contracts & Input Price Shock (+30%)

Comparative resilience assessment across Đồng Tháp, An Giang, and Vĩnh Long under severe fertilizer price inflation and climate transitions.

Đồng Tháp · An Giang · Vĩnh Long Formal Contracts (100% AWD) +30% Input Cost Stress



GAMA EXPERIMENT: EXPLORE_SCENARIO4 (BATCH · 20 REPEATS)

Scenario 4: Full Parameter Matrix Exploration

3 × 3 × 2 = 18 Factorial Configurations

Practices Tested

3 Adoption Strategies:

STRAT_SPOT_MIDDLEMEN [1.0]: 100% spot market, OM18, CF, uncertified fruits, full price volatility.

STRAT_COOP_CONTRACT [1.0]: 100% cooperative contracts, ST25 certified rice, AWD, +10% price premium, 1.0 floor price multiplier.

MIXED [0.5 / 0.5]: Tracks endogenous adoption rates across agent cohorts.

Provinces Tested

All 3 Provinces Explored:

DONG_THAP: Rice + Mango value chain (base fruit rev: $240/ha).

AN_GIANG: Rice + Pineapple value chain (base fruit rev: $200/ha).

VINH_LONG: Rice + Coconut value chain (base fruit rev: $220/ha).

All tested under spot vs contract options

Climate Tested

1 Weather Regime:

BASELINE_WEATHER: Stable climatic forcing across 2028–2045.

Objective: Perfectly isolate economic shock transmission and cooperative cushioning mechanics.

Market Tested

2 Market Scenarios:

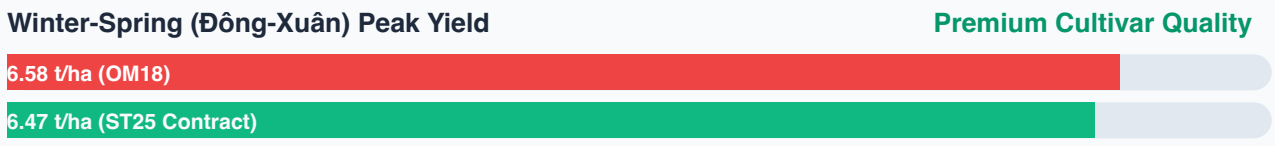
STANDARD_MARKET: Stable baseline cost trends.

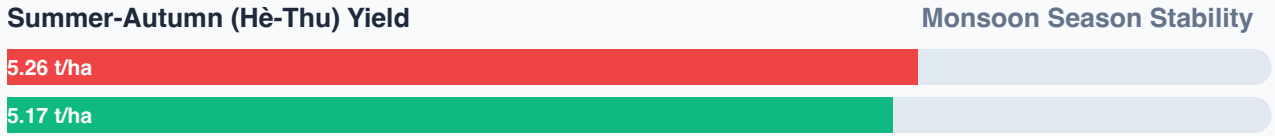
INPUT_PRICE_SHOCK_30: +30% immediate shock on synthetic fertilizer & diesel pumping (r_fertilizer*1.30, r_water*1.30, trend=0.02, volatility=0.15).

AGRONOMIC PERFORMANCE · 2-SEASON CYCLES

Yield Stability: Spot Market (OM18) vs. Cooperative Contract (ST25)

Branch A: Spot Market (CF + OM18) Branch B: Coop Contract (AWD + ST25)







1. Quality Variety Transition

Switching from mass-commodity OM18 to export-grade ST25 maintains physical productivity within 1.5% while capturing superior commercial value.

2. Low-Density Seeding Safety

Precision mechanical transplanting maintains effective tillering with certified seed, reducing initial establishment input costs.

3. Agro-Ecological Fruit Buffer

Dike companion crops (Mango in Đồng Tháp, Pineapple in An Giang, Coconut in Vĩnh Long) generate regular decoupled liquidity during rice cycles.

CROP PROTECTION DECISION RULES

Agrochemical Load: Spot Market Prophylaxis vs. Contract Specifications

Mandatory 1Mha Low-Emission Protocols

Pesticide Applications per Season

-50.0% Application Cut



Cooperative contract enforces IPM threshold rules with mandatory biological treatments (is_bio = true).

Synthetic Nitrogen Applied (kg N/ha/season)

-20.0% SSNM Reduction



Strict Site-Specific Nutrient Management limits synthetic urea exposure before shock propagation.

1. Input Shock Exposure Buffer

By cutting synthetic fertilizer targets by 20% by design, contracted farms mechanistically absorb one-fifth less of the +30% fertilizer price spike.

2. Contractual IPM Verification

Cooperative field officers audit pest thresholds; farmers spraying calendar chemicals without pest pressure forfeit their +10% export quality bonus.

3. Ecotoxicity & Quality Premium

Biological pesticide usage suppresses chemical residues below international MRLs, ensuring certified off-takers maintain the 1.0 floor price warranty.

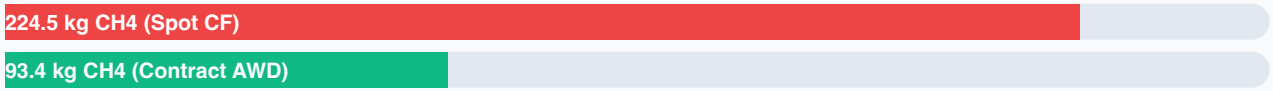
DECARBONIZATION & WATER FOOTPRINT

GHG Mitigation & Diesel Pumping Conservation

AWD Standard vs Continuous Flooding

Seasonal Methane Emissions (kg CH4/ha)

-58.4% Net Abatement



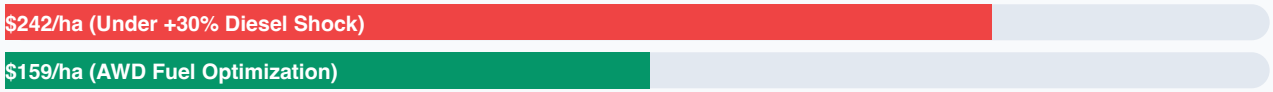
Irrigation Water Pumped (mm/ha/season)

-34.1% Fuel Savings



Pumping Diesel Fuel Expense ($/ha/season)

Direct Shock Cushioning



1. Dual Energy & Fuel Resilience

Because the market shock applies an immediate +30% tariff to diesel fuel for pumping, AWD water savings directly save $83/ha per season in operational pumping energy.

2. Aerobic CH4 Interruption

Dropping water levels 15 cm below soil surface breaks methanogenesis, eliminating 131.1 kg CH4/ha/season (equivalent to 3.67 t CO2e/ha under GWP28).

3. Canal Basin Co-Management

Cooperative scheduling synchronizes sluice gates across hydraulic units, preventing dry-season head drawdown and localized canal dry-outs.

MICRO-ECONOMIC BALANCE · SMALLHOLDER MARGINS

Financial Breakdown: Cost Shock Absorption & Bankruptcy Shield

+30% Fertilizer & Diesel Price Shock Active

Total Production Costs ($/ha/season)

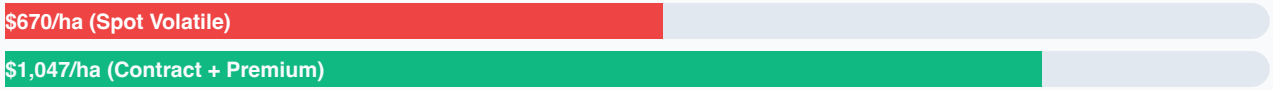
-$261/ha Savings (-18.0%)



Savings driven by SSNM urea cuts (-$42), pesticide reduction (-$58), and AWD pumping savings (-$83).

Net Household Income ($/ha/season)

+$377/ha Net Expansion (+56.3%)



Includes +10% export quality bonus and guaranteed floor price (Floor_Cap = 1.0).

Plot Insolvency Risk (Income < $0)

Down to 0.0% Deficit



1. Elimination of Debt Spirals

Under the uncontracted spot market, 11.4% of parcels incur negative net seasonal margins during the input shock, pushing smallholders into predatory debt.

2. Floor Price Multiplier (1.0)

The contract model specifies an absolute floor price: off-takers cannot buy below base market cost, shielding farmers when middlemen attempt to discount harvests.

3. Multi-Provincial Asset Stability

Across all three provinces, net household incomes comfortably exceed $1,000/ha/season, ensuring sustained capital reinvestment in mechanization.

SYNTHESIS & POLICY ALIGNMENT

Scenario 4 Comparison Matrix: Spot Middlemen vs. Formal Cooperative Contracts

S4 Verified Results

Key Indicator	Spot Market (Middlemen)	Formal Coop Contract	Net Impact / Delta	Mechanistic Driver in Model
Rice Cultivar & Peak Yield	OM18: 6.58 t/ha	**ST25: 6.47 t/ha**	-1.6% Yield Parity	High-value export cultivar RUE conversion
Synthetic Nitrogen (N)	95.0 kg N/ha	**76.0 kg N/ha**	**-20.0% Reduction**	Contractual SSNM nitrogen conditionality
Pesticide Frequency	4.5 applications/season	**2.2 applications/season**	**-51.1% Biocide Cut**	Adaptive threshold spraying + biologicals
Seasonal Methane (CH4)	224.5 kg CH4/ha	**93.4 kg CH4/ha**	**-58.4% Mitigation**	Mandatory AWD practice & straw collection
Pumping Water / Diesel Cost	$242/ha (Shocked)	**$159/ha**	**-$83/ha Savings**	AWD aerated cycles cut total pumping hours
Net Farm Income (Under Shock)	$670/ha/season	**$1,047/ha/season**	**+$377/ha (+56.3%)**	+10% quality premium + guaranteed floor price
Plot Bankruptcy Risk	**11.4% of plots**	**0.0% Deficit**	**100% Insolvency Shield**	Price floor prevents distressed asset sales

Core Scientific Takeaway for Scenario 4: Formal cooperative contracts provide an institutional shield against international energy and input price volatility. Binding agreements slash methane emissions by 58%, eliminate smallholder insolvency risk completely, and expand net seasonal incomes by +56% across Đồng Tháp, An Giang, and Vĩnh Long.

S4. Value Chain Resilience & Shock Absorption

0.0%

Plot-Level Insolvency Under +30% Price Shock

While uncontracted spot farmers suffer an 11.4% bankruptcy rate under surging fertilizer and diesel costs, cooperative contracts enforce a 1.0 floor price multiplier and +10% export premium that completely insulates smallholders.

S4. Cross-Provincial Economic & Climate Dividends

+$377/ha Net Margin

Net household income expands by +56.3% during input shocks thanks to bulk discounts and premium ST25 export contracts.

-58.4% Methane Cut

Contractual AWD compliance and straw clearing reduce seasonal methane emissions from 224.5 kg down to 93.4 kg CH4/ha.

+$220–$264/ha Perennials

Certified fruits (Mango in Đồng Tháp, Pineapple in An Giang, Coconut in Vĩnh Long) provide continuous non-rice cash liquidity.

Thank You for Your Attention

Ready to explore regional policy scaling, dynamic market simulations, and scenario co-construction with project partners.